Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**4**
Electrons (Ne)	**4**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**4**
Eigenstate idx	**0 (ground state)**

Eigenstate properties

Energy E0	**-2.10274848**
Energy E1	**-1.80642385**
Gap (E1-E0)	**0.296325**
<S^2>	**0.000000**
S	**0.0000**
<Sz>	**0.000000**
Multiplicity	**1.0**
Spin label	**Singlet (S=0)**

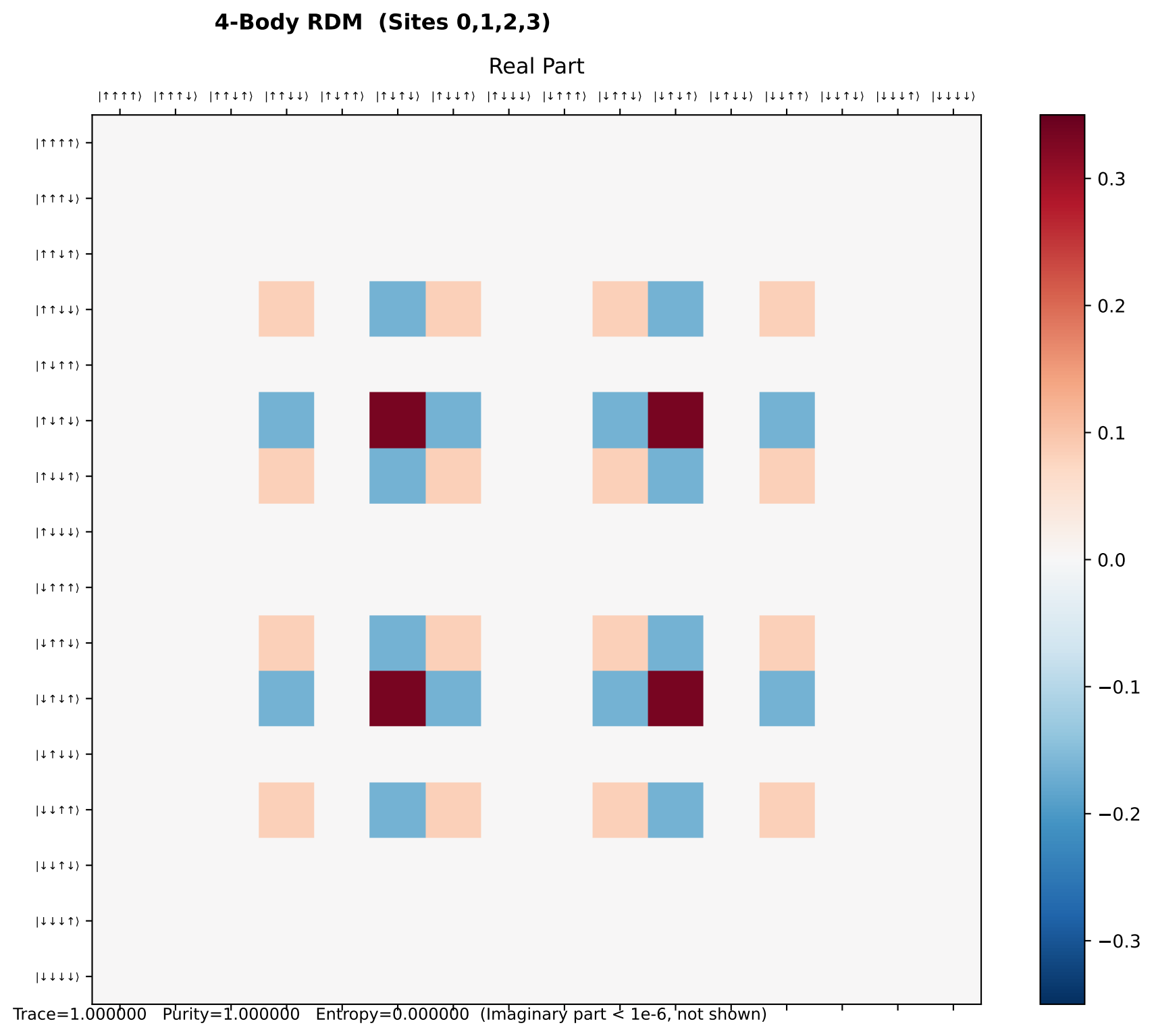
Double occupancy

<D> total	**0.287325**
<D> / site	**0.071831**

RDM and GME statistics for this run (4-body clusters)

Hilbert space dim : 2^4 = 16
Number of clusters : 4
Pure clusters : 4
Mixed clusters : 0
Clusters with GME (>0) : 4

Purity gamma = Tr(rho^2) : mean = 1.000000 [min 1.000000 , max 1.000000]
Von Neumann entropy : mean = 0.000000 [min 0.000000 , max 0.000000]
Rank of rho : mean = 1.00 [min 1 , max 1]
C_GME (best estimate) : mean = 0.912871 [min 0.912871 , max 0.912871]


4-Body RDM (Sites 0,1,2,3)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,2,3)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -4.749088e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (0,)         | (1, 2, 3)    | 0.50000000
  (0, 1)       | (2, 3)       | 0.58333333
  (0, 2)       | (1, 3)       | 0.33333333
  (0, 3)       | (1, 2)       | 0.58333333
  (0, 1, 2)    | (3,)         | 0.50000000
  (0, 1, 3)    | (2,)         | 0.50000000
  (0, 2, 3)    | (1,)         | 0.50000000

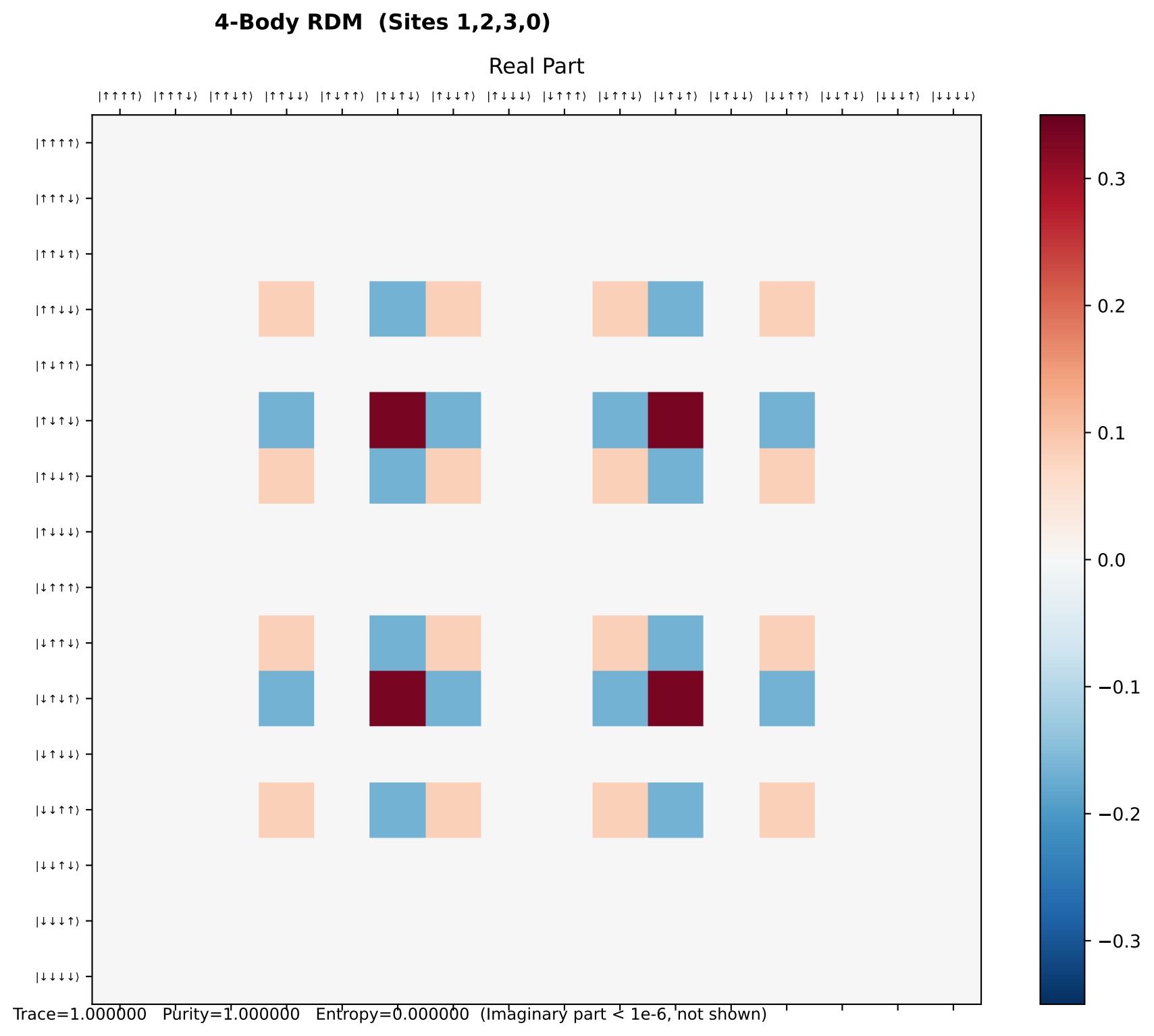
STATUS: GME EXISTS
------------------------------------

C_GME = 0.91287093
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
==================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.91287093   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.91287093

  Status: GME
```


4-Body RDM (Sites 1,2,3,0)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,2,3,0)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -7.045364e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (1,)         | (2, 3, 0)    | 0.50000000
  (1, 2)       | (3, 0)       | 0.58333333
  (1, 3)       | (2, 0)       | 0.33333333
  (1, 0)       | (2, 3)       | 0.58333333
  (1, 2, 3)    | (0,)         | 0.50000000
  (1, 2, 0)    | (3,)         | 0.50000000
  (1, 3, 0)    | (2,)         | 0.50000000

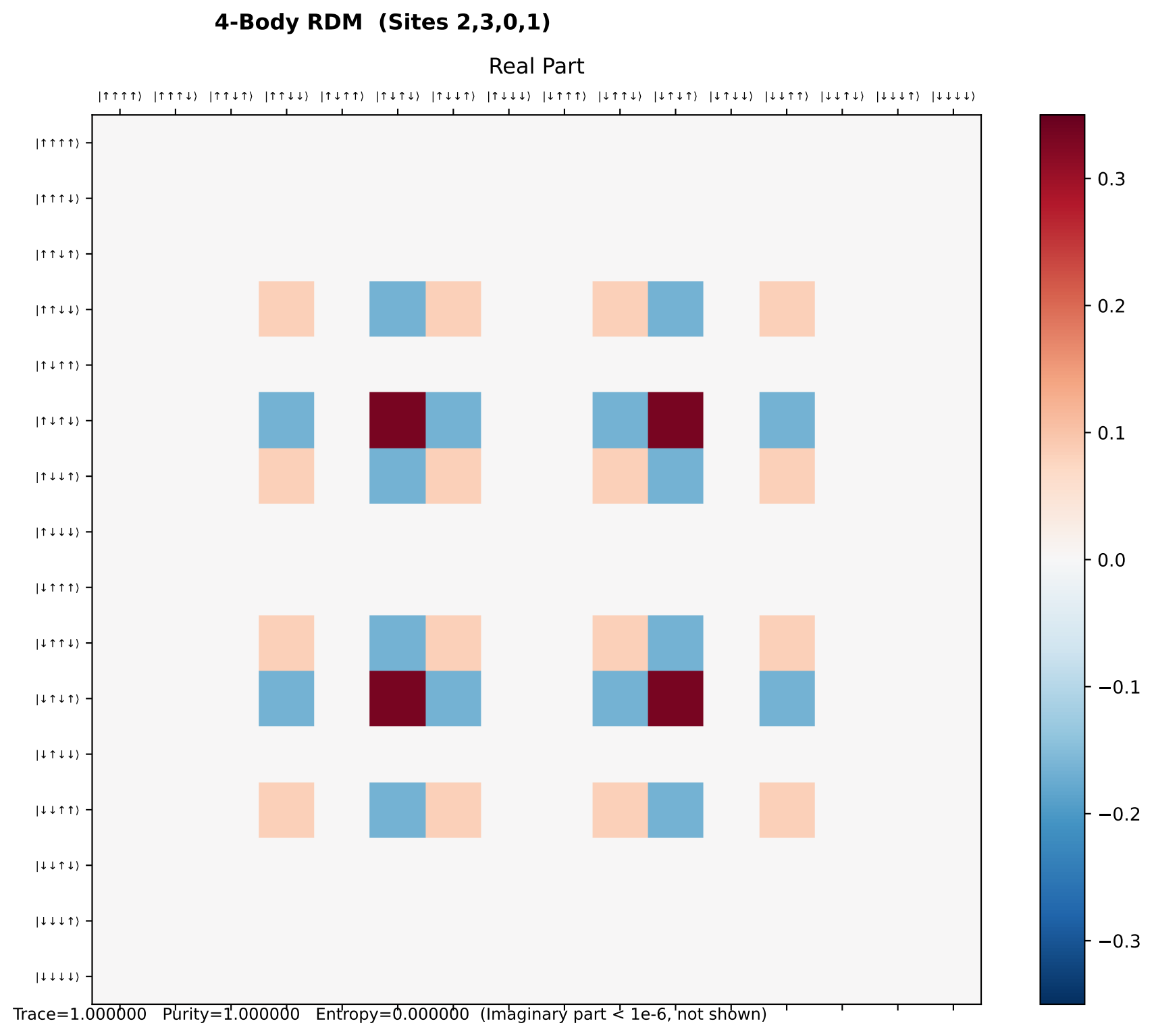
STATUS: GME EXISTS
------------------------------------

C_GME = 0.91287093
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
==================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.91287093   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.91287093

  Status: GME
```


4-Body RDM (Sites 2,3,0,1)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 2,3,0,1)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -1.068409e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (2,)         | (3, 0, 1)    | 0.50000000
  (2, 3)       | (0, 1)       | 0.58333333
  (2, 0)       | (3, 1)       | 0.33333333
  (2, 1)       | (3, 0)       | 0.58333333
  (2, 3, 0)    | (1,)         | 0.50000000
  (2, 3, 1)    | (0,)         | 0.50000000
  (2, 0, 1)    | (3,)         | 0.50000000

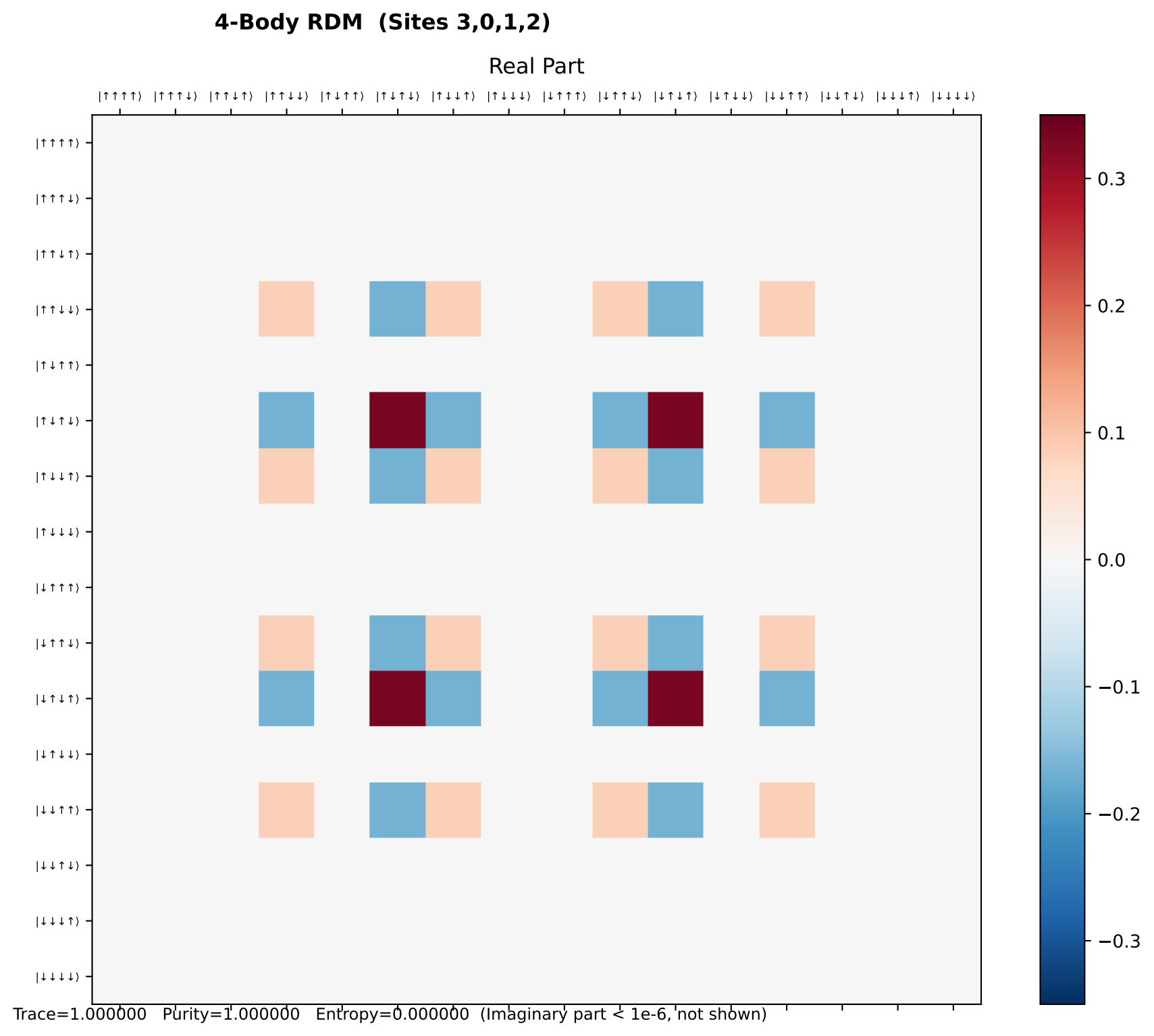
STATUS: GME EXISTS
------------------------------------

C_GME = 0.91287093
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.91287093   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.91287093

  Status: GME
```


4-Body RDM (Sites 3,0,1,2)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 3,0,1,2)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -9.909913e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (3,)         | (0, 1, 2)    | 0.50000000
  (3, 0)       | (1, 2)       | 0.58333333
  (3, 1)       | (0, 2)       | 0.33333333
  (3, 2)       | (0, 1)       | 0.58333333
  (3, 0, 1)    | (2,)         | 0.50000000
  (3, 0, 2)    | (1,)         | 0.50000000
  (3, 1, 2)    | (0,)         | 0.50000000

STATUS: GME EXISTS
------------------------------------

C_GME = 0.91287093
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
==================================

  Lower Bounds:
    C_LB    = 0.00000000
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  Upper Bound:
    C_UB (eigendecomp)  = 0.91287093

  Status: GME
```